Customer Churn Prediction

Model Evaluation and Comparison

Based on the evaluation results you provided, here is a comparison of the two models:

Model	Accuracy	Precision (Class 1)	Recall (Class 1)	F1-Score (Class 1)
SVM	0.8610	0.83	0.40	0.54
KNN (K=5)	0.8240	0.62	0.34	0.44

Observations

- SVM achieves a higher overall accuracy (86.1% vs 82.4%) and significantly better precision for the churned class (0.83 vs 0.62). This means that when SVM predicts a customer will churn, it is correct 83% of the time, whereas KNN is correct only 62% of the time.
- However, both models have low recall for churned customers (0.40 and 0.34), meaning they miss a large portion of actual churners (60% and 66% false negatives).
- The F1-score (harmonic mean of precision and recall) is higher for SVM (0.54) than for KNN (0.44), indicating SVM

is the better model overall for the minority class.

Business Implication

If the goal is to identify as many potential churners as possible (high recall), neither model performs well; I would need to address class imbalance (e.g., using SMOTE or class weights). If the goal is to be confident when marking a customer as "churn" (high precision) – perhaps to allocate retention resources efficiently – SVM is the better choice.

Conclusion

We developed two classification models for predicting customer churn:

- **Support Vector Machine (SVM)** with default RBF kernel.
- **K-Nearest Neighbors (KNN)** with optimal K=5 selected via the elbow method.

Based on my provided evaluation metrics, SVM outperforms KNN on accuracy, precision for churned customers, and F1-score. Therefore, for this dataset and split, SVM is the

preferred model.

To further improve performance, especially recall for the churned class, I could:

- Use techniques to handle class imbalance (e.g., oversampling the minority class, adjusting class weights).
- Perform hyperparameter tuning for both models (e.g., grid search for C and gamma in SVM, different distance metrics in KNN).
- Engineer additional features from the existing data (e.g., balance-to-salary ratio, product utilization, etc.).